

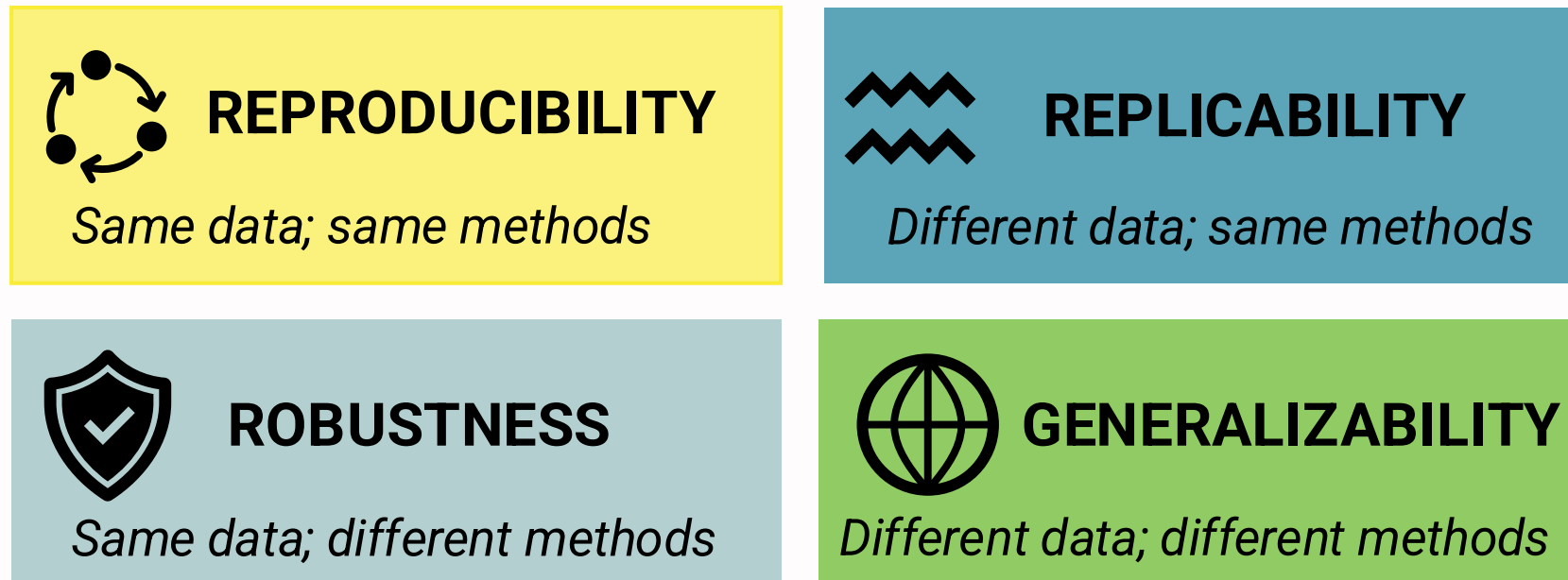
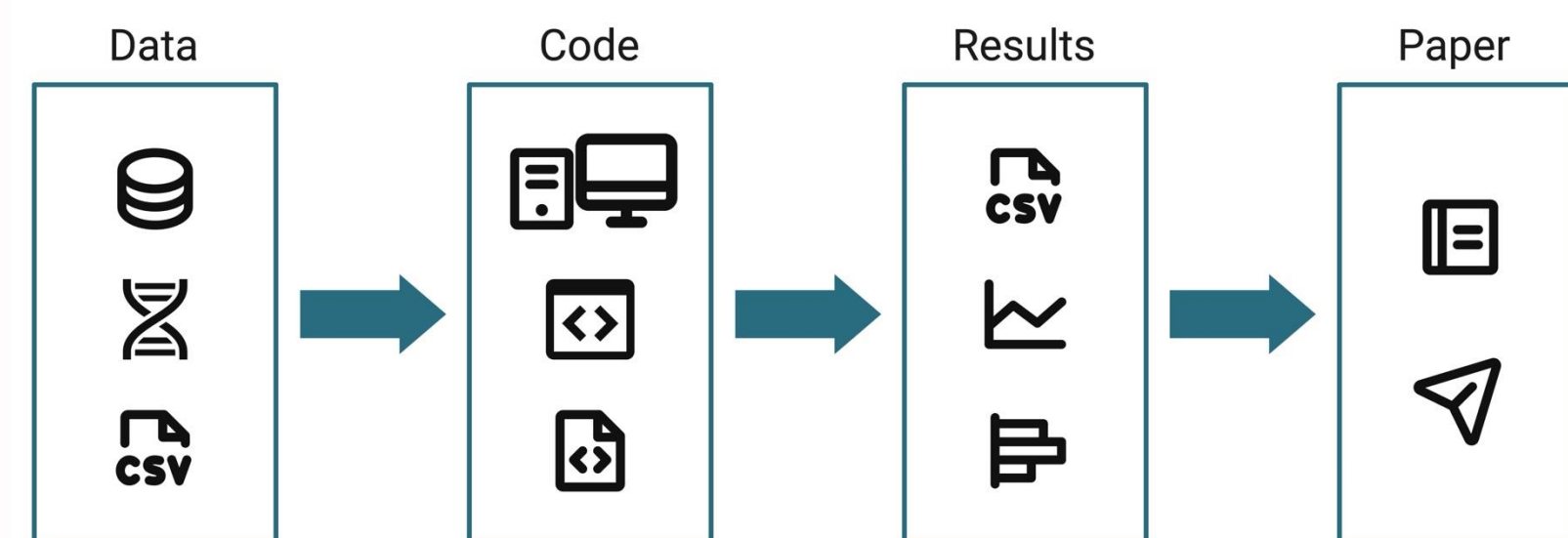
Anatomy Of A Modular Toolchain For Reproducible Bioinformatics Workflows

Kelly Sovacool^{1,2}, Samarth Mathur^{1,2}, Vishal Koparde^{1,2}, Parthav Jailwala^{1,2}

1 Advanced Biomedical Computational Science, Frederick National Laboratory for Cancer Research, Frederick, MD 21702, USA
2 CCR Collaborative Bioinformatics Resource, Office of Science and Technology Resources, National Cancer Institute, National Institutes of Health, Bethesda, MD, USA

INTRODUCTION

Reproducibility is the foundation of trustworthy computational biology. Our framework implements core reproducible principles all pipelines.



Schloss PD. 2018. 10.1128/mBio.00525-18.

BEST PRACTICES

Version Control

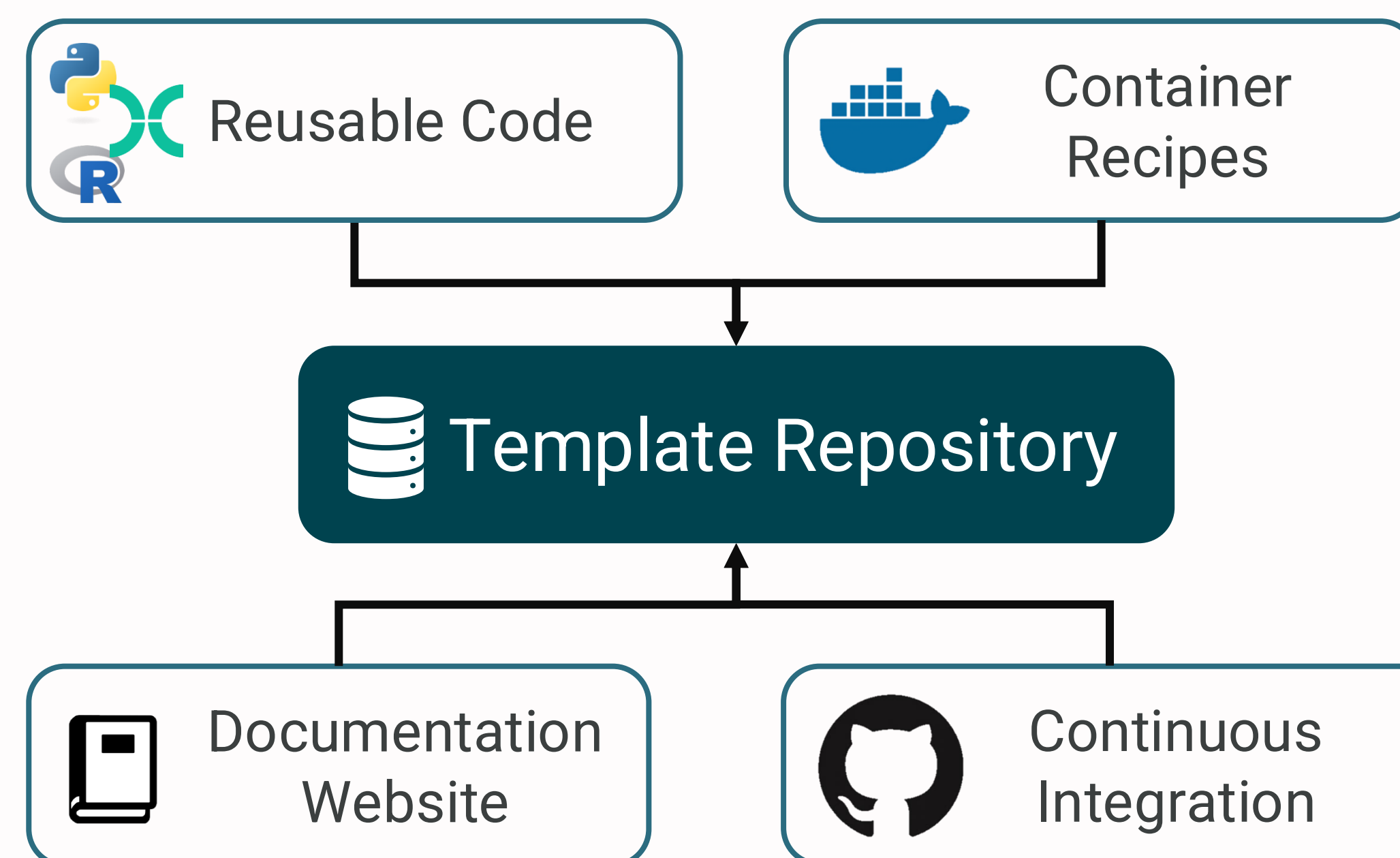
Modular Design

Automated Tests

Documentation

We enforce consistency through templates, modularity, and automation to prevent technical debt and improve maintainability.

TOOLCHAIN FOR REPRODUCIBLE BIOINFORMATICS



Unified framework from template to deployment with automated quality checks.

Fix once,
deploy everywhere

Consistent CLI

Automated testing
& quality checks

PIPELINES BUILT

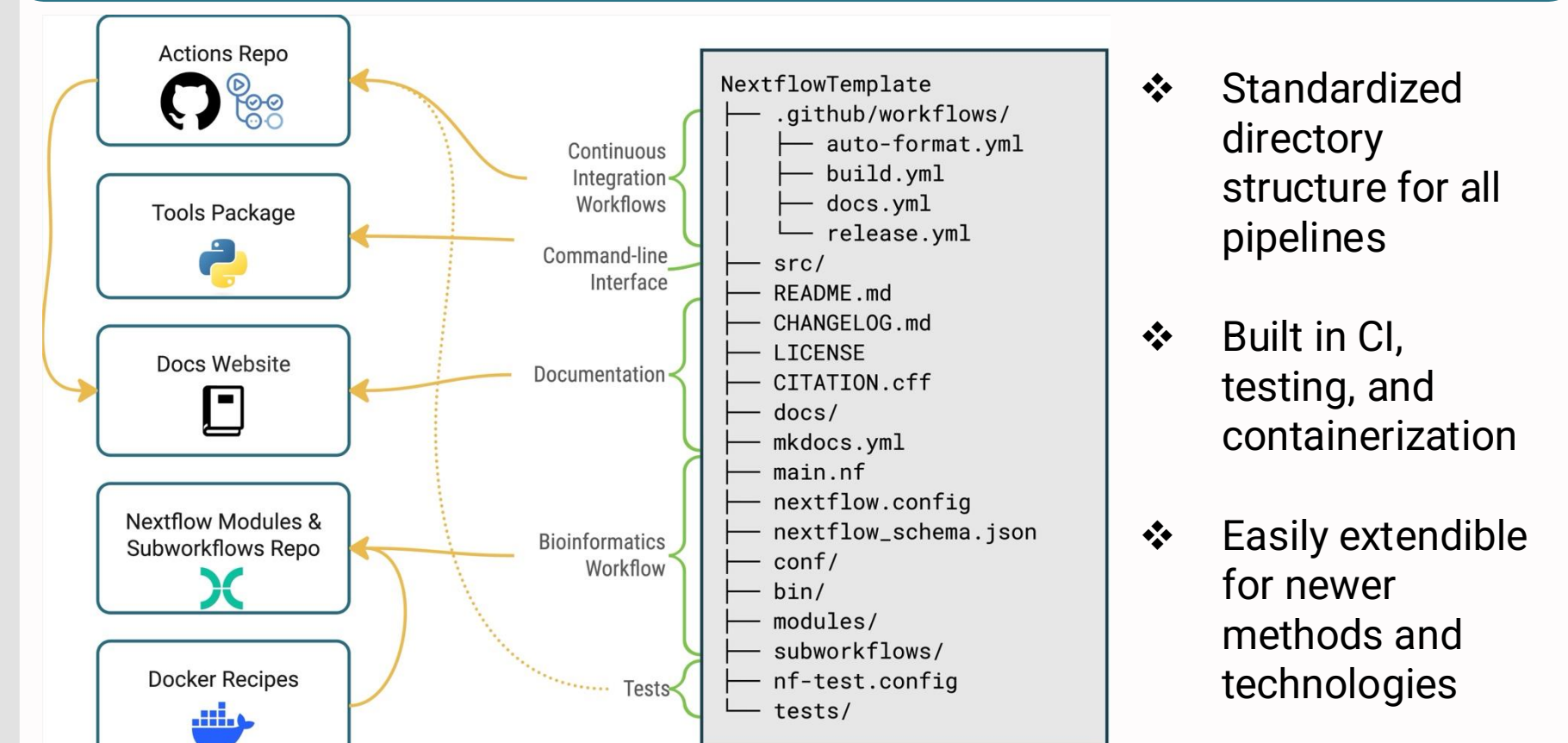


10+

pipelines on NIH Biowulf HPC cluster

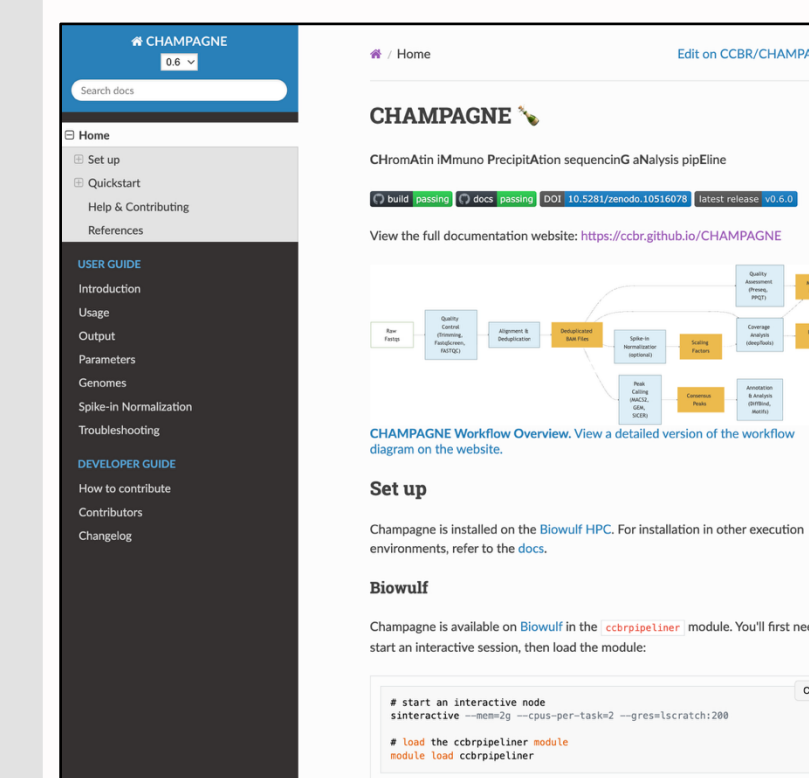
\$ module load ccbripipeline

PIPELINE TEMPLATE



- ❖ Standardized directory structure for all pipelines
- ❖ Built in CI, testing, and containerization
- ❖ Easily extendible for newer methods and technologies

CLI, DOCUMENTATION, AUTOMATION



```
$ champagne run --input samplesheet
$ sinclair run --input samplesheet
```

Consistent usage guidelines and documentation across all pipelines simplifies user training.

GitHub Actions runs CI checks on every change.

Scalable, Reproducible, Maintainable Bioinformatics

NIH Biowulf HPC

Secure, Scalable, and high performance computing platform

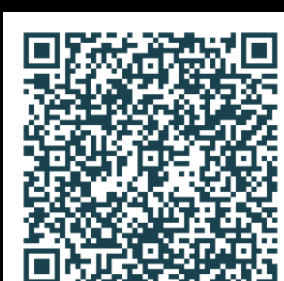
GitHub Open Source

Collaborate, Contribute, Reuse

Documentation

Guides, Tutorials, Best Practices

This Poster



Template Repo



Advancing reproducible computational research through standardized, containerized, and automated bioinformatics pipelines

This project has been funded in whole or in part with Federal funds from the National Cancer Institute, National Institutes of Health, under Contract No. 75N91019D00024. The content of this publication does not necessarily reflect the views or policies of the Department of Health and Human Services, nor does mention of trade names, commercial products, or organizations imply endorsement by the U.S. Government.

Frederick National Laboratory is a Federally Funded Research and Development Center operated by Leidos Biomedical Research, Inc., for the National Cancer Institute